



A comparison of nutrient-based and exchange-group methods of diet instruction for patients with noninsulin-dependent diabetes¹⁻³

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ABSTRACT This study compared effectiveness of nutrient-based (Diet Guide) vs food-group (Exchange Lists) methods of diabetic diet evaluation in improving dietary compliance, glycemic control, and biochemical indicators of heart disease risk. Eighty-three persons with noninsulin-dependent diabetes were taught one of two diet-planning methods in a 3-session workshop. Both methods led to reductions in energy intake and percent of calories from fat and saturated fatty acids in 6 mo postworkshop. Reductions in fat intake were greater and more long lasting for persons using Diet Guide than using Exchange Lists method of diet planning. Despite dietary changes, neither diet-planning method led to significant decreases in weight or skinfold thickness. Few differences were seen in clinical measurements pre- and 6 mo postworkshop. Total and LDL cholesterol values were lower than preworkshop values for men in both groups. Suggestions are given for improving effectiveness of both diet-planning methods. *Am J Clin Nutr* 1987;45:625-37.

KEY WORDS Instruction methods, dietary intake, anthropometric data, clinical data, diabetes, nutrient-based diet, exchange lists, nutrition education methods

Introduction

Vascular disease is a major long-term complication of diabetes, accounting for an estimated 75% of all deaths among known diabetics (1). In response to a growing concern about the importance of dietary intervention aimed at reducing risk factors associated with coronary heart disease, the Committee on Food and Nutrition of the American Diabetes Association in 1980 revised principles of diet therapy for persons with diabetes in accordance with general recommendations for reducing risk of atherosclerosis (2). Recommendations included adjustment of energy intake such that 12-20% is derived from protein, 50-60% from carbohydrates, up to 10% from polyunsaturated fatty acids, < 10% from saturated fatty acids, and the remainder from monounsaturated fatty acids. The Committee also recommended a modest restriction in sodium intake and increase in consumption of foods rich in fiber. Education of persons with diabetes and their families was seen as key in

developing and maintaining effective treatment regimens. Allowance for flexibility and individuality in diet and meal planning were considered of major importance in achieving dietary adherence.

Unfortunately, it has been found repeatedly that persons with diabetes rarely understand or follow their diet prescriptions (3-8). Adherence has been better when multifaceted and expanded nutritional care programs have been implemented (9-14).

Currently the most common diet-planning method used by physicians and dietitians in the United States with diabetic patients is the

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Exchange Lists for Meal Planning (15), a food group approach in which common foods are grouped into six lists. It was suggested by several researchers that traditional food group approaches may be less effective in promoting proper nutrition and diet than desired (16–18). The National Institute of Health consensus development panel has recommended community development research be conducted to test the effectiveness of various approaches to nutrition education designed to influence food choices (19).

A nutrient approach has been recommended as an effective way to teach nutrition (20–24). With the development of specific recommendations on energy-supplying nutrients for diabetics and others at increased risk of heart disease, a nutrient approach can be used in planning diets to meet specific goals for macronutrients as well as for vitamins and minerals.

This study was designed to compare the effectiveness of a nutrient-based diet evaluation method, the Colorado State University *CSU Diet Guide for Planning Prudent Diets* (25), with that of the traditional diabetic diet evaluation method, the *ADA Exchange Lists for Meal Planning* (15), in improving dietary compliance among persons with noninsulin-dependent diabetes (NIDDM) as evidenced by dietary self-report, anthropometric, and biochemical measures.

Materials and methods

Experimental design

Eighty-three persons with NIDDM (57 female, 26 male) completed the study. The protocol of the study was approved by the Committee on Human Research for Colorado State University. Prospective patients were recruited from physician and hospital lists in the Fort Collins, Colorado, area. Through use of a randomized block design, subjects were classified by sex and preference for afternoon or evening sessions. They were then ordered by percentage mean desirable weight for height (% MDW) and randomly assigned as matched pairs in descending order of % MDW to the Diet Guide (DG) or the Exchange List (EL) diet planning treatment group. The recruitment procedure was conducted twice so that in all, diet-planning workshops were presented to four DG and four EL groups. To prevent instructor bias in patient education, a local dietitian assisted the senior author with the first DG and EL workshops presented per phase and presented the second DG and EL workshops per phase.

Educational programs

Materials developed for the DG educational program include the *CSU Diet Guide for Planning Prudent Diets*

(25), worksheets for planning and evaluating menus, a Leader's Guide, and three slide-cassette tape programs on 1) diabetes, 2) using the Diet Guide, and 3) expanding the Diet Guide. The Diet Guide is divided into 18 sections. The first two sections provide information on diet and chronic disease and on using the manual in menu planning and evaluation. The last 14 sections are devoted to food composition data for ~ 1800 menu items. Information is presented in standardized units called nutri-units to simplify dietary evaluation. Total kilocalories, percent of calories from macronutrients, cholesterol, fiber, sodium, and five vitamins and two minerals are included in the evaluation.

The EL educational program was designed to resemble in structure and content the DG program. The major difference was that the *Exchange Lists for Meal Planning* (15) was used as the menu-planning and evaluation tool. A slide-cassette tape program, *Meal Planning with Exchange Lists* (Coal MS, Smith WM, Spall KE; 80 slides, 22-min cassette tape; Medical Univ of South Carolina, 1979), was used to help teach the Exchange Lists.

Subjects were taught basic principles of nutrition and diet for diabetes and how to use the diet-planning method, to which they had been randomly assigned, in three 2½-h workshops held at 1 wk intervals. A similar format was used for teaching both diet-planning methods. Additional details on materials developed and on structure and format of both workshop series are presented elsewhere (26, 27).

Dietary evaluation

Three-day dietary records were obtained from each subject before the first workshop and at the 3-mo and 6-mo follow-up sessions. All subjects received detailed instructions at enrollment on how to complete their dietary records. Records were reviewed for completeness by the data analyst and missing details were filled in during an interview with the subject. Food records were reviewed and blind coded for computer processing and analysis so that data analyzers were not aware of treatment group.

Nutrient analysis was completed using CSU's computer program NUTCAL II. This data file is based on the 1982 USDA standard nutrient data base release (28) supplemented with nutrient data on zinc and total sugar (reducing sugars plus sucrose) and augmented with information from manufacturers on fortified cereals and from large, franchised restaurants on fast foods. Nutrients analyzed included total kilocalories, protein, total fat, saturated fatty acids, polyunsaturated fatty acids, cholesterol, carbohydrate, total sugar, crude fiber, vitamin A, vitamin C, thiamin, riboflavin, niacin, calcium, phosphorus, iron, sodium, and zinc. Dietary intakes were expressed as amounts per day and as percentages of the Recommended Dietary Allowances (RDA) of the Food and Nutrition Board (FNB) (29).

Anthropometric measurements

Height, weight, triceps skinfold, midarm circumference, and blood pressure were measured at the time of enrollment and at the 3- and 6-mo follow-up sessions. Height and weight were measured in light clothing without shoes and were used with Metropolitan Life Insurance Company Desirable Weight Tables (30) to calculate mean desirable weight for height (MDW), relative weight (% MDW), and Quetelet index (weight/height² in g/cm²).

Midarm circumference (MAC) and triceps skinfold (TSF) were measured midway between the tip of the acromion and the olecranon process of the nondominant arm by the same technician. MAC was measured to the nearest 0.5 cm using a plastic tape, while the nondominant arm hung relaxed at the subject's side. TSF was measured 1 cm above MAC with the arm held pendant. Measurements were read after 3 s of caliper application with a Lange Skinfold Caliper (Cambridge Scientific Industries, Cambridge, MD) having a pressure of 10 g/mm² of contact surface area. Triplicate measurements of each TSF were taken and the mean was used for statistical analyses and anthropometric calculations. From MAC and TSF measurements, upper-arm fat area (F), in mm², was calculated using the equation of Frisancho (31).

Blood pressure was measured with a standard mercury sphygmomanometer. Measurements were performed by the same health nurse with the subject comfortably seated. The mean value of two readings taken 10 min apart was used for analysis.

Biochemical measurements

Fasting plasma glucose, glycohemoglobin (HbA_{1c}), cholesterol, HDL cholesterol, and triglycerides were measured just before the workshops and 6-mo postworkshop. Fasting plasma glucose was measured on the Beckman Glucose Analyzer by the Beckman-developed oxygen rate method (Beckman Instruments, Fullerton, CA). Total glycosylated hemoglobin (HbA_{1c}) was determined by the Bio-Rad Hemoglobin A_{1c} column test method (Bio-Rad Clinical Division, Richmond, CA). Total serum cholesterol was measured by the Technicon SMAC Cholesterol (Enzymatic-II) method (Technicon Instruments Corp, Tarrytown, NY). HDL cholesterol was determined according to the method of Burnstein et al (32) as modified by Lopes-Virella et al (33). Serum triglycerides were analyzed on the Technicon SMAC Biochemical Analyzer by the enzymatic method of Bucolo and David (34). LDL cholesterol was calculated from total and HDL cholesterol and triglyceride values by the following equation: LDL cholesterol = cholesterol - [HDL cholesterol + (triglycerides/5)].

Statistical methods

The study used a repeated-measures design with subjects nested within treatment group and sex across time. Results were analyzed by analysis of variance (ANOVA) with Statistical Package for the Social Sciences (SPSS) programs (McGraw-Hill, New York). When a main effect (treatment group, sex, or time) with two or more categories or an interaction among main effects was significant, the Student-Newman-Keuls procedure (35) was used to determine level of significance.

Results

Subjects

Subject characteristics are presented in Table 1. Significant differences in these characteristics did not occur between treatment groups. Participant age ranged from 21 to 82 yr. Nearly 90%, however, were over 40. Years

TABLE 1
Characteristics of subjects completing study

	Diet guide group (n = 42)	Exchange list group (n = 41)
Men	14	12
Women	28	29
Age (yr)	56.2 ± 16.0*	60.2 ± 13.8
Education (yr)	13.6 ± 3.5	12.3 ± 2.9
Length of diagnosis (months)	69.9 ± 76.9	80.9 ± 81.1
Using insulin	7	7
Using oral hypoglycemic agents	11	16

* Mean ± SD.

of schooling completed ranged from 3 yr elementary school to > 4 yr of college; 78% were high school graduates and 26% were college graduates. Length of diagnosis of diabetes ranged from < 1 mo to > 25 yr; 34.2% had been diagnosed as diabetic for < 1 yr, 18.9% for 1–3 yr, and 30.3% for > 10 yr. Approximately one-third (32.5%) were on oral hypoglycemic agents and 16.2% were on insulin.

Participants were asked several questions about their diets at the enrollment meeting. Approximately three-fourths reported that they had been given a diet prescription by their physician or dietitian. Only 47%, however, indicated they were still following the diet. Of the others, approximately one-fourth indicated they were not following their diet and one-fourth indicated that they sometimes followed their diet. Of the 22 participants who did not complete the study, 17 (77%) indicated that they had experienced difficulty following a diet prescription.

Energy, protein, fat, carbohydrate, and fiber intake

Dietary energy, distribution of energy intake from macronutrients, and intake of cholesterol and crude fiber before the diet-planning workshops and 3 and 6 mo postworkshop are presented in Table 2. Initial intakes of these nutrients did not vary between treatment groups. Men consumed significantly more calories per day ($p < 0.001$) than did women. Men also consumed a greater percentage of calories as fats and saturated fatty acids ($p < 0.05$) and more mg cholesterol ($p < 0.05$). In contrast, women consumed a greater percentage of calories as carbohydrates and sugars ($p < 0.05$).

TABLE 2

Energy, protein, fat, carbohydrate, and fiber intake of diet-planning treatment groups over time*

	Diet guide			Exchange list			SEM†	Main effects and significant interactions‡		
	Male (n = 14)	Female (n = 28)	Total (n = 42)	Male (n = 12)	Female (n = 29)	Total (n = 41)		Group	Sex	Time
Kilocalories										
Pre	2241	1464	1797	2328	1578	1953	± 64.6	NS	p < 0.001	p < 0.01
3 mo	1862	1268	1523	1896	1323	1609	± 56.0			
6 mo	1679	1251	1434	1865	1332	1598	± 46.9			
Protein, % of kcal										
Pre	18.8	18.3	18.5	17.8	17.1	17.4	± 0.3	NS	NS	< 0.05
3 mo	18.3	20.6	19.6	17.2	19.1	18.2	± 0.3		time × sex	p < 0.05
6 mo	20.2	21.2	20.7	18.1	18.5	18.3	± 0.4			
Total fat, % of kcal										
Pre	42.6	39.1	40.6	48.9	38.7	43.8	± 0.8	NS	p < 0.05	p < 0.05
3 mo	41.2	34.6	37.4	44.7	35.1	39.9	± 0.9		time × group	p < 0.05
6 mo	38.7	33.4	35.7	46.4	36.9	41.6	± 0.8			
Saturated fatty acids, % of kcal										
Pre	15.6	13.5	14.4	19.7	14.0	16.8	± 0.4	NS	p < 0.05	p < 0.05
3 mo	15.2	12.3	13.5	17.7	12.3	15.0	± 0.4			
6 mo	14.5	11.5	12.8	18.8	12.7	15.8	± 0.4			
Polyunsaturated fatty acids, % of kcal										
Pre	6.5	6.7	6.6	6.3	5.5	5.9	± 0.3	NS	NS	NS
3 mo	6.1	5.5	5.8	5.2	5.8	5.5	± 0.3			
6 mo	5.4	5.5	5.5	5.9	5.8	5.8	± 0.3			
Cholesterol, mg										
Pre	468.8	285.3	364.0	616.3	346.2	481.8	± 18.8	NS	p < 0.05	NS
3 mo	389.7	266.6	319.4	418.7	267.9	343.3	± 17.6			
6 mo	394.2	274.9	326.1	460.7	307.8	384.3	± 17.2			
Carbohydrate, % of kcal										
Pre	38.5	43.7	41.5	33.4	44.8	39.1	± 0.8	NS	p < 0.05	NS
3 mo	40.5	46.6	43.9	38.5	46.9	42.7	± 0.9			
6 mo	40.6	46.9	44.2	35.1	45.2	40.2	± 0.8			
Sugar, % of kcal										
Pre	16.7	19.1	18.1	11.3	19.6	15.4	± 0.6	NS	p < 0.05	NS
3 mo	15.4	19.8	17.9	13.6	20.0	16.8	± 0.6			
6 mo	15.6	20.8	18.6	13.5	19.0	16.3	± 0.6			
Crude fiber (g/1000 kcal)										
Pre	2.4	3.0	2.8	2.1	3.1	2.6	± 0.2	NS	NS	NS
3 mo	2.9	3.3	3.1	2.4	3.3	2.9	± 0.2			
6 mo	2.8	3.3	3.1	2.1	3.0	2.5	± 0.2			

* Adjusted means of 3-day food records preworkshop, 3 mo postworkshop, and 6 mo postworkshop.

† Standard error of measurement of pooled means.

‡ p values for analysis of variance; NS, not significant.

Total energy intake decreased ($p < 0.01$) in both groups in the 6 mo following diet-education intervention. No significant differences were seen in the magnitude of the decrease between DG and EL treatment groups or between men and women. All of the energy-

supplying nutrients contributed to the decline, but the greatest impact came from declines in intake of total fat and saturated fatty acids. Total fat intake as well as percentage of calories from fat dropped significantly ($p < 0.01$) in both groups in the first 3 mo postworkshop.

Intake continued to decrease ($p < 0.05$) from the 3- to 6-mo postworkshop evaluation for DG subjects but not for EL subjects. At the 6-mo evaluation, fat supplied 35.7% of total calories (57.7 g) for DG compared with 41.6% (76.4 g) for EL subjects. As with total fat, actual intake and percentage of calories from saturated fat declined ($p < 0.05$) in both treatment groups between the preworkshop and the 3-mo postworkshop evaluations. Intake continued to decline between the 3- and 6-mo postworkshop evaluations among DG subjects compared with slight increases in mean intake among EL subjects; differences at this point, however, were not significant. Intake of polyunsaturated fatty acids also declined ($p < 0.05$) in both treatment groups in the first 3 mo postworkshop. However, because of an equivalent decrease in total calories, no differences in percent of calories from polyunsaturated fatty acids were seen.

Initial intake of saturated and polyunsaturated fatty acids resulted in a polyunsaturated-to-saturated fatty acid ratio (P:S) of 0.48 for DG and 0.38 for EL subjects, both well under the 1.0 ratio recommended by the American Diabetes Association for persons at risk of heart disease. Because of decreases in both saturated and polyunsaturated fatty acids in the months following the workshop, P:S ratio did not change with time nor were any significant differences seen between DG and EL subjects or between male and female subjects at any time.

Although mean intake of protein decreased slightly in the 6 mo postworkshop, percentage of calories from protein showed a significant increase ($p < 0.05$). Differences in magnitude of the increase were not seen between treatment groups. Women as a group, however, showed greater increases in the percent of calories from protein than did males ($p < 0.05$). Dietary cholesterol declined for all groups; crude-fiber intake (g/1000 kcal) increased slightly for DG subjects and remained steady for EL subjects in the 6 mo postworkshop. However, for both nutrients, differences seen were not significant (Table 2).

Percent recommended intakes of energy, protein, vitamins, and minerals

Dietary intakes over time of energy, protein, five vitamins, and four minerals for male and

female subjects in the two treatment groups, expressed as percentage of 1980 RDA, are given in Table 3. For each subgroup studied, mean energy intake initially was below RDA and declined further ($p < 0.01$) in the 6 mo postworkshop. Significant differences in the magnitude of the decline did not occur between DG and EL treatment groups or between men and women.

Despite the low energy intake, mean intakes of protein and all five vitamins evaluated (vitamin A, vitamin C, riboflavin, thiamin, and niacin) were above RDA for all subgroups, both before and after the diet-education programs. No significant differences existed between DG and EL groups or between men and women as a result of educational intervention, possibly because initial levels were already above recommended levels.

Calcium intake was initially higher ($p < 0.01$) for men (116% RDA) than for women (86% RDA). In the 6 mo postworkshop, calcium intake decreased for DG men ($p < 0.05$) but not for EL men. In contrast, intakes increased slightly for DG women and decreased for EL women. As a result, calcium intake at 6 mo postworkshop was higher ($p < 0.05$) for DG women than for EL women (87.7% vs 75.0% RDA) and lower ($p < 0.01$) for DG men than for EL men (95.4% vs 123.4% RDA).

Initial mean iron intake (% RDA) was greater ($p < 0.01$) for men than for women and above the RDA for all subgroups (172.9% and 114.0% of RDA for men and women, respectively). Consistent with decreases in energy consumption postworkshop, mean intake of iron declined for all subgroups, but differences were not significant. At the end of the study, mean intake was still well above recommended level for men and close to recommended level for women (141.0% and 97.1% RDA, respectively). Most women were postmenopausal with a lower RDA for iron than for the few premenopausal women in the study. The mean intake of iron among women at 6 mo postworkshop (10.9 mg) was well below recommended levels for premenopausal women.

Preworkshop zinc intake was considered adequate for men (109.9% RDA) and marginal for women (64.0% RDA). Consistent with decreasing energy intake, zinc intake declined ($p < 0.01$) in the 6 mo postworkshop for both diet-planning groups.

TABLE 3
Energy and nutrient intakes of diet-planning treatment groups over time expressed as percent of RDA*

Nutrient % RDA	Diet guide						Exchange lists						ANOVA†			
	Male (n = 14)			Female (n = 28)			Male (n = 12)			Female (n = 29)			Group	Sex	Time	
	pre	3 mo	6 mo	pre	3 mo	6 mo	pre	3 mo	6 mo	pre	3 mo	6 mo				
Kcal	92.1	75.4	68.6	79.6	69.3	68.2	95.8	77.5	76.5	85.7	72.1	72.2	± 2.0	NS	NS	< 0.01
Protein	183.1	146.7	148.4	153.4	144.8	146.8	183.5	143.4	148.8	148.8	142.2	137.2	± 3.7	NS	NS	NS
Vitamin A	160.7	144.4	136.0	160.8	181.5	174.2	134.9	130.7	162.4	176.4	152.2	152.8	± 8.4	NS	NS	NS
Vitamin C	215.9	201.5	236.3	179.2	171.6	178.2	169.4	167.1	155.2	194.1	214.4	177.7	± 10.5	NS	NS	NS
Thiamin	121.0	122.9	108.1	114.2	104.6	114.6	145.4	121.9	123.8	126.3	113.7	112.8	± 4.4	NS	NS	NS
Riboflavin	139.7	128.4	119.5	130.0	124.6	131.0	157.2	117.6	142.3	141.5	117.3	113.8	± 4.2	NS	NS	NS
Niacin	169.2	144.4	137.5	130.9	135.5	139.0	161.5	118.0	129.5	144.9	124.3	123.3	± 4.1	NS	NS	NS
Calcium	117.2	97.1	95.4	85.7	80.8	87.7	115.1	109.9	123.4	86.4	80.4	75.0	± 3.9	NS	NS	NS§
Phosphorus	201.2	169.4	159.3	139.1	130.5	132.4	189.4	165.5	184.9	141.1	130.5	120.8	± 4.8	NS	< 0.01	NS
Iron	177.9	166.4	147.9	103.4	98.5	97.7	169.1	143.4	136.0	124.6	106.0	96.5	± 5.3	NS	< 0.01	NS
Zinc	130.8	75.2	69.9	60.9	56.3	54.3	94.3	78.4	78.3	67.0	55.6	49.8	± 2.7	NS	< 0.05	< 0.05

* Adjusted means of 3-day food records before workshop, 3 mo postworkshop, and 6 mo postworkshop.

† Standard error of measurement of pooled means.

‡ Analysis of variance for main effects; NS, not significant.

§ p < 0.05 interaction among group, sex, and time for calcium.

Sodium and potassium

Dietary intakes of sodium and potassium before and 3 and 6 mo postworkshop are presented in Table 4. Initial intakes were similar between treatment groups and higher ($p < 0.01$) for men than women. For both elements initial values were within estimated safe and adequate ranges recommended by FNB (29), with the exception of sodium intake for male EL subjects. Initial sodium intakes also were consistently above levels often recommended for persons at increased risk of hypertension and heart disease (36).

Declines in sodium intake were noted for male and female DG and male EL subjects in the postworkshop evaluations, but individual variation was high and none of the differences were significant. At the 6-mo postworkshop evaluation, mean sodium intake for female DG subjects was 1840 mg/day, well below the maximum level of 2000 mg/day recommended in the *CSU Diet Guide* for persons at risk of hypertension.

Clinical data

Anthropometric measurements and blood pressure of the two diet-planning treatment groups pre- and postworkshop are presented in Table 5. Mean weight, relative weight, and Quetelet index did not vary between the DG and EL treatment groups at enrollment, 3 mo postworkshop, or 6 mo postworkshop. Men weighed significantly more ($p < 0.01$) but were closer to their mean desirable weight ($p < 0.05$)

than women were at each evaluation. All three weight measurements showed slight declines in the 6 mo postworkshop, but differences were not significant for the group as a whole, by treatment group (DG vs EL), or by sex.

Triceps skinfold and upper-arm fat area did not differ between DG and EL treatment groups but were significantly higher ($p < 0.01$) for women than for men at each evaluation. Although small declines in the two measurements were seen in both treatment groups in the 6 mo following the programs, especially for male EL subjects, differences were not significant.

Initial blood pressure values did not vary between treatment groups or sexes. For each subgroup mean levels were $< 140/90$ mmHg level used to define elevated blood pressure in the HANES I study (37). Significant decreases in systolic blood pressure were not seen for either diet planning treatment group at the time of the 3- and 6-mo postworkshop evaluations. Six-month postworkshop diastolic pressure, however, was lower ($p < 0.05$) than preworkshop levels for male DG and female EL subjects.

Biochemical measurements are presented in Table 6. Initial mean plasma glucose levels of DG and EL subjects (164 mg/dL and 144 mg/dL, respectively), were well above the normal range of 65–110 mg/dL. At 6 mo postworkshop, mean plasma glucose for DG subjects had decreased to 148 mg/dL and for EL subjects had remained steady at 144 mg/dL.

TABLE 4
Intake of sodium and potassium as influenced by diet-planning method*

Group	n	Mean daily intake					
		Sodium (mg)			Potassium (mg)		
		Pre	3 mo	6 mo	Pre	3 mo	6 mo
Diet guide							
Male†	14	3195	3026	2458	3462	2843	2673
Female	28	2321	2045	1840	2517	2260	2285
Total	42	2696	2466	2105	2923	2510	2451
Exchange list							
Male†	12	4363	3086	2663	2978	2709	2662
Female	29	2407	2215	2605	2439	2347	2240
Total	41	3385	2650	2634	2708	2528	2451
SEM‡		± 122.8	± 100.3	± 195.5	± 96.1	± 80.4	± 76.3

* Adjusted means of 3-day food records before workshop, 3 mo postworkshop, and 6 mo postworkshop.

† Male intake > female intake ($p < 0.01$) for both sodium and potassium.

‡ Standard error of measurement of pooled means.

TABLE 5
Anthropometric measurements and blood pressure of diet-planning treatment groups over time*

Measurement	Diet guide			Exchange list			SEM†	Main effects and significant interactions‡		
	Male (n = 14)	Female (n = 28)	Total (n = 42)	Male (n = 12)	Female (n = 29)	Total (n = 41)		Group	Sex	Time
Body weight, lb (kg)										
Pre	182.5 (83.0)	164.7 (74.9)	172.4 (78.4)	180.7 (82.1)	162.0 (73.6)	171.3 (77.9)	± 3.8 (± 1.7)	NS	p < 0.01	NS
3 mo	181.2 (82.4)	164.8 (74.9)	171.8 (78.1)	181.4 (82.5)	159.8 (72.6)	170.6 (77.5)	± 3.8 (± 1.7)			
6 mo	179.7 (81.7)	164.4 (74.7)	171.0 (77.7)	180.0 (81.8)	158.7 (72.1)	169.4 (77.0)	± 3.8 (± 1.7)			
Relative weight, % MDW										
Pre	117.6	136.5	128.4	117.1	133.9	125.5	± 0.3	NS	p < 0.05	NS
3 mo	116.9	136.6	128.2	117.5	132.1	124.84	± 0.3			
6 mo	116.0	136.3	127.6	116.8	131.2	124.08	± 0.3			
Quetelet index										
Pre	0.27	0.29	0.28	0.27	0.28	0.28	± 0.05	NS	NS	NS
3 mo	0.26	0.29	0.28	0.27	0.28	0.27	± 0.05			
6 mo	0.26	0.29	0.28	0.26	0.28	0.27	± 0.05			
Triceps skinfold, mm										
Pre	15.9	28.9	23.3	17.2	29.0	23.1	± 1.1	NS	p < 0.001	NS
3 mo	16.6	28.1	23.2	15.4	27.7	21.6	± 1.1			
6 mo	15.6	27.4	22.3	14.1	27.4	20.7	± 1.1			
Upper arm fat area, mm²										
Pre	2278	4150	3348	2458	4087	3273	± 181.7	NS	p < 0.01	NS
3 mo	2394	4059	3345	2250	3896	3073	± 181.9			
6 mo	2304	3952	3246	2047	3881	2964	± 180.6			
Systole, mmHg										
Pre	131	133	132	127	131	129	± 1.9	NS	NS	NS
3 mo	127	130	129	130	131	131	± 1.7			
6 mo	126	130	128	128	125	126	± 1.7			
Diastole, mmHg										
Pre	80	78	79	78	78	78	± 0.8	NS	NS	NS
3 mo	78	81	79	83	78	80	± 1.0			
6 mo	77	80	78	79	75	77	± 1.0			

* Adjusted means of measurements taken preworkshop, 3 mo postworkshop, and 6 mo postworkshop.

† Standard error of measurement of pooled means.

‡ Analysis of variance; NS, not significant.

However, variations in scores within groups were so great that no significant effect on plasma glucose levels over time was attributable to the diet-planning method used. Likewise, differences in mean plasma glucose level were not seen between men and women initially or 6 mo postworkshop.

In contrast with plasma glucose, initial mean HbA_{1c} levels (8.1% Hb for both groups) were only slightly above the upper end of the normal range of 4.0–8.0% Hb. Within groups, initial HbA_{1c} levels were higher (p < 0.05) for

DG men and EL women (8.8 and 8.5% HbA_{1c}, respectively). This pattern persisted at the 6-mo postworkshop examination. Significant effects on HbA_{1c} 6 mo postworkshop were not seen for either group.

Initial mean serum cholesterol levels did not vary between treatment groups or sexes. Although declines were seen with all subgroups in the 6 mo postworkshop, the magnitude of the decline was significant (p < 0.05) only for men. Six months postworkshop, serum cholesterol levels had declined from 224 mg/dL

TABLE 6
Plasma glucose, glycosylated hemoglobin, and serum lipid levels of diet-planning treatment groups over time*

Measurement	Diet guide			Exchange list			SEM†	Main effects and significant interaction‡		
	Male (n = 14)	Female (n = 28)	Total (n = 42)	Male (n = 12)	Female (n = 29)	Total (n = 41)		Group	Sex	Time
Plasma glucose, mg/dL										
Pre	179	153	164	138	150	144	± 6.5	NS	NS	NS
6 mo	152	144	148	141	147	144	± 5.5			
Hemoglobin A _{1c} , %										
Pre	8.8	7.5	8.1	7.8	8.5	8.1	± 0.2	NS	NS	NS
6 mo	8.9	8.1	8.4	8.3	8.8	8.6	± 0.2	group × sex (p < 0.05)		
Serum cholesterol, mg/dL										
Pre	212	219	216	232	222	227	± 5.0	NS	NS	NS
6 mo	196	217	208	218	215	217	± 5.0	time × sex (p < 0.05)		
LDL-cholesterol, mg/dL										
Pre	143	138	140	159	139	149	± 4.0	NS	NS	NS
6 mo	126	135	131	143	132	138	± 4.4	time × sex (p < 0.01)		
HDL-cholesterol, mg/dL										
Pre	39	44	42	41	47	44	± 1.5	NS	p < 0.01	NS
6 mo	41	44	43	42	49	46	± 1.3			
Serum triglycerides, mg/dL										
Pre	152	190	174	160	180	170	± 12.8	NS	NS	NS
6 mo	144	189	170	162	168	165	± 12.8			

* Adjusted means of measurements taken preworkshop and 6 mo postworkshop.

† Standard error of measurement of pooled means.

‡ Analysis of variance; NS, not significant.

to 208 mg/dL for men, compared with a decline from 221 to 216 mg/dL for women.

Preworkshop mean serum LDL cholesterol levels were similar for the two treatment groups and higher (p < 0.01) for men than women (152 vs 138 mg/dL, respectively). Consistent declines were noted in LDL cholesterol for all subgroups at 6 mo postworkshop but again were significant (p < 0.01) only for men. Serum LDL cholesterol levels at 6 mo postworkshop were 136 mg/dL for men and 134 mg/dL for women.

Initial mean HDL cholesterol levels were similar for treatment groups and higher (p < 0.01) for women than men (45 vs 40 mg/dL, respectively). HDL cholesterol increased slightly in the 6 mo postworkshop for all subgroups, but differences were not significant. Mean serum triglyceride levels did not vary between treatment groups or as a function of sex preworkshop or 6-mo postworkshop.

Discussion

Both diet-planning methods led to significant decreases (p < 0.01) in energy intake in the first 3 mo postworkshop. Energy intake of DG subjects continued to decline between the 3- and 6-mo postworkshop follow-up but leveled off for EL subjects. However, for both treatment groups subsequent decreases in body weight were slight and not statistically significant.

Reasons for this are not clear. Food records are an indirect measure of dietary intake and, therefore, are prone to reporting errors. Lansky and Brownell (38) reported this especially to be the case for dieting obese patients and questioned the utility of food records for estimating energy intake or predicting weight loss.

On the other hand, studies evaluating dietary assessment methods have reported 3-day

food records to be reasonable estimates of the general quality of the diet (39–41). Gersovitz et al (39) found 3- and 4-day records less likely to be prone to the flat-slope syndrome (under-reporting high intakes and over-reporting low intakes) than 24-h recalls and less likely to be biased due to dropouts and decreased record usability than 7-day records.

Even if 3-day records are a reasonable estimate of the general quality of the diet, it is possible that subjects adjusted their diets on days they kept diet records. Indeed, self-monitoring of food intake has been shown to be an effective treatment procedure for short-term weight loss (42). However, it also is possible that the subjects did reduce their energy intake in the 6 mo following the workshops, but because of a corresponding decrease in energy expenditure or resting metabolism, the body was able to maintain its weight in the face of decreased energy intake. It has long been known that resting metabolism drops in response to calorie restriction (43, 44) and that weight loss seldom equals the restriction in calories (45). Further, since obesity tends to increase with age despite observed decreases in energy intake (46), it is possible, as seen in the Stanford Three Community Study (47), that the very small but steady decrease in mean weight represented a reversal of what might have been an increase in body weight over the study period.

Changes in energy intake following the workshops resulted in a lower percent of total calories supplied as fat and saturated fatty acids and greater percent as carbohydrates and protein for both groups in the first 3 mo post-workshop. Percent of calories supplied as fat and saturated fatty acids continued to decrease and percent supplied as carbohydrates and protein to increase between the 3- and 6-mo follow-up sessions for DG but not EL subjects. As a result of these trends, at 6 mo postworkshop, DG subjects were consuming 35.7% of their calories as fat and 12.8% as saturated fatty acids, compared with 41.6% and 15.8%, respectively, by EL subjects. Although both groups were still above levels recommended by RDA, DG subjects showed greater progress in achieving and maintaining the recommended goals than did EL subjects. Both workshops and diet-planning methods stressed a low-fat diet, but the DG method of diet

planning provided more specific evaluative information on the total and saturated fatty acid content of the daily diet. It is possible that this added information provided the awareness, knowledge, and motivation needed by DG subjects to continue to strive toward a low-fat diet long after the motivation provided by the workshop series itself had worn off. This significant reduction in fat consumption by DG participants occurred 6 mo after workshops with a total instructional time of < 7 h.

In addition to evaluating energy intake from macronutrients, DG subjects evaluated daily intake of three vitamins and two minerals. Mean intakes for each of these were at or above recommended levels for all subgroups both before and after the workshops except for calcium. Initial calcium intake was adequate for men (116% RDA) and marginal for women (86% RDA). In the 6 mo postworkshop, calcium intake decreased for EL women, consistent with overall decrease in food energy. In comparison, intake for DG women increased despite decreases in total kilocalories, reflecting an effort by DG women to include calcium-rich foods in their diets. As a result, at the end of 6 mo DG women were consuming 88% RDA for calcium compared with 75% for EL women. This difference is noteworthy, especially considering the attention calcium has received in recent years in prevention and treatment of osteoporosis among postmenopausal women (48, 49) and the recent link between low levels of calcium intake and increased incidence of hypertension (50).

Three other nutrients discussed in the workshops of both diet-planning groups but specifically evaluated with the DG method were crude fiber, cholesterol, and sodium. At the beginning of the study, DG subjects were consuming 67% and EL subjects 61% of the recommended 7.5 g/day crude fiber (25). Crude fiber intake did not change significantly in the 6 mo postworkshop for either group. Many subjects indicated difficulty increasing crude-fiber intake. Since fiber, especially soluble types, is becoming increasingly recognized as an important food for controlling glycemia (51, 52), this is an area that might benefit from increased attention in future workshops for diabetics. Because of incomplete information on the dietary fiber content of foods at the time of this study, data pre-

sented in Table 2 evaluated only the crude fiber content of foods. This may not give a true picture of dietary-fiber intake of the subjects or of changes made as a result of the workshops, especially for soluble fibers such as legumes and oat bran, stressed in the workshops but inadequately evaluated in the analysis of crude fiber. (The *Diet Guide* has since been updated so that it includes information on dietary fiber.)

Initial intakes of cholesterol and sodium were close to the DG goals of < 300 mg cholesterol and < 2000 mg sodium for women but not for men. For both nutrients, intake decreased among DG and EL subjects in the 6 mo postworkshop. However, variations in consumption were so large that neither decreases seen nor differences in dietary changes between DG and EL subjects were statistically significant. Considering the already low intake of these nutrients and the closeness of mean intakes at 6 mo postworkshop to DG goals, the lack of significance in the findings is not unexpected or disappointing.

Changes in blood glucose and lipid levels at 6 mo postworkshop were in the direction desired but significant at the $p < 0.05$ level only for declines in serum cholesterol and LDL cholesterol for men. Significant decreases in total and LDL cholesterol have been reported in a number of studies on diabetic and non-diabetic subjects for whom intake of total fat, saturated fatty acids, and cholesterol were reduced (53, 54). In this study, initial reports of dietary cholesterol and percent of calories from fat and saturated fatty acids were closer to recommended levels for women than for men. Therefore, one might expect greater response to dietary changes by men.

In contrast to Reaven (55), carbohydrate-induced hypertriglyceridemia was not observed in this study. Rather, serum triglyceride levels were slightly lower at 6 mo postworkshop. Similar results have been reported in other studies (56, 57).

In evaluating the results, several points need to be kept in mind. The subjects were predominantly elderly persons who were not newly diagnosed diabetic. Some had been diagnosed for 25 yr; mean length of diagnosis was 6.2 yr. Many appeared complacent with their current eating patterns and not highly motivated to make dietary changes as long as

their symptoms did not become worse. This is understandable when one considers, as pointed out by Eckerling and Kohrs (8), that adhering to a diabetic diet is a reminder that one is ill; when symptoms are not present, it is easier to forget the disease and the therapeutic regimen. Total instructional time with either method was only 7 h with no follow-up measurement of blood glucose or lipid levels until 6 mo postworkshop. Results suggest that more instructional time with greater emphasis on changing behavior, more family support, more follow-up sessions in which blood glucose and lipids were measured, and perhaps use of home glucose monitoring would have been helpful in promoting the desired changes. Success in obtaining desired dietary changes has been better in those studies involving frequent follow-up sessions (8–14).

In comparing response to the two diet-planning methods it is important to remember that the EL group was not a usual care group. The EL method consisted of three well-developed workshops to teach the Exchange Group method. This is more diet education than is often given to diabetic patients (4–7). Also, most DG subjects had previously been exposed to the EL method of diet planning. Therefore, differences in response to the two diet-planning methods might be expected to be small.

On the other hand, there are a number of very positive findings of this study that should be stressed. The DG method developed is new and is a complete departure in concept from the well-established exchange-group method that has been used for many years. It is less rigid in menu development aspects and more quantitative in the evaluation of foods consumed in relation to goals. In spite of this, a study population averaging > 60 yr of age not only was able to master the program but used it with great enthusiasm and interest. As discussed elsewhere (26), nutrition knowledge and attitude toward dietary management of diabetes improved significantly in the process. The audience, because of age and length of time with diagnosed diabetes, was not the easiest one to work with and, yet, the method was favorably received. Preliminary data from a follow-up study shows that > 75% of the subjects have continued to use the Diet Guide in the 2 yr following the initial workshops (Heller K, personal communication). It is anticipated

that newly diagnosed patients in the 40–60 yr age group might have used the DG method even more effectively. A few subjects had access to a microcomputer and said they would have appreciated a computerized version of the Diet Guide. Such a program compatible with the IBM-PC has since been developed. The computerized version (PRUCAL) includes riboflavin, vitamin B-6, and zinc as indicator nutrients and replaces crude-fiber with dietary-fiber values. (Information on the program is available from the authors.)

Diet education is becoming an increasingly important field. As no single approach is likely to meet the needs of all people, additional work is needed to determine those patient characteristics most likely to benefit from a nutrient approach vs those most likely to benefit from a food group approach to diet education. ■

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